

LAB EXPERIMENT – 2

Implementing Data Versioning Using DVC

Course Outcome: CO1

Aim

To understand how DVC versions datasets in Machine Learning projects.

Step 1: Create Another Notebook

Open

Google Colab → File → New Notebook → Experiment_2_DVC.ipynb

Step 2: Mount Google Drive

```
from google.colab import drive
```

```
drive.mount('/content/drive')
```

Click

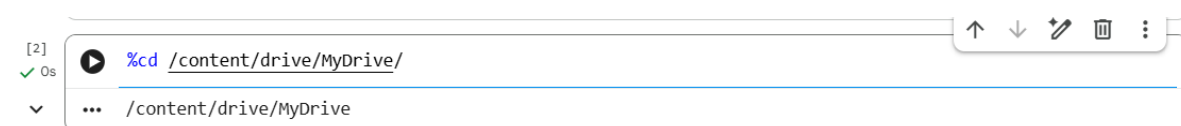
Allow Access



The screenshot shows a Google Colab notebook cell with the following code: `from google.colab import drive` and `drive.mount('/content/drive')`. The cell has been executed successfully, indicated by a green checkmark and the text "[1] ✓ 29s". Below the code, the output shows "... Mounted at /content/drive". The left sidebar of the Colab interface is visible, showing icons for file explorer, search, and other notebook functions.

Step 3: Move to Google Drive

```
%cd /content/drive/MyDrive/
```



The screenshot shows a Google Colab notebook cell with the command `%cd /content/drive/MyDrive/`. The cell has been executed successfully, indicated by a green checkmark and the text "[2] ✓ 0s". Below the code, the output shows "... /content/drive/MyDrive". The right sidebar of the Colab interface is visible, showing icons for undo, redo, and other notebook functions.

Step 4: Create Project Folder

```
!mkdir DVC_Project
```

Move into the folder

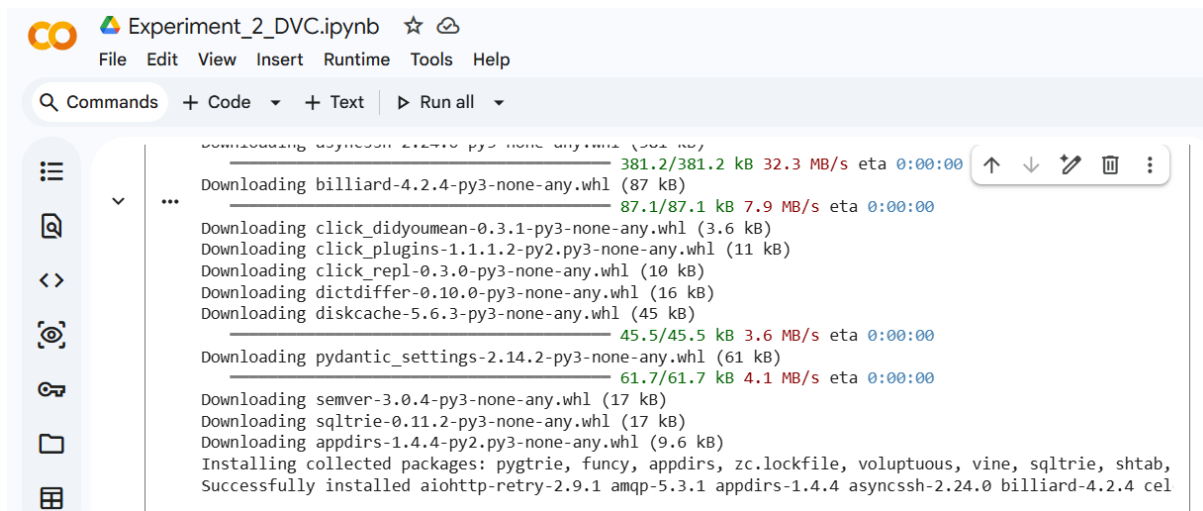
```
%cd DVC_Project
```



The screenshot shows two Google Colab notebook cells. The first cell contains the command `!mkdir DVC_Project` and has been executed successfully, indicated by a green checkmark and the text "[3] ✓ 0s". The second cell contains the command `%cd DVC_Project` and has also been executed successfully, indicated by a green checkmark and the text "[4] ✓ 0s". Below the code, the output shows "... /content/drive/MyDrive/DVC_Project". The right sidebar of the Colab interface is visible, showing icons for undo, redo, and other notebook functions.

Step 5: Install DVC

`!pip install dvc`



The screenshot shows a Jupyter Notebook titled "Experiment_2_DVC.ipynb". The interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help) and a toolbar with options like "Commands", "Code", "Text", and "Run all". The left sidebar contains icons for file management and execution. The main cell displays the output of a `!pip install dvc` command, showing the progress of downloading various packages. The output is as follows:

```
...
Downloading billiard-4.2.4-py3-none-any.whl (87 kB)
381.2/381.2 kB 32.3 MB/s eta 0:00:00
Downloading click_didyoumean-0.3.1-py3-none-any.whl (3.6 kB)
87.1/87.1 kB 7.9 MB/s eta 0:00:00
Downloading click_plugins-1.1.1.2-py2.py3-none-any.whl (11 kB)
Downloading click_repl-0.3.0-py3-none-any.whl (10 kB)
Downloading dictdiffer-0.10.0-py3-none-any.whl (16 kB)
Downloading diskcache-5.6.3-py3-none-any.whl (45 kB)
45.5/45.5 kB 3.6 MB/s eta 0:00:00
Downloading pydantic_settings-2.14.2-py3-none-any.whl (61 kB)
61.7/61.7 kB 4.1 MB/s eta 0:00:00
Downloading semver-3.0.4-py3-none-any.whl (17 kB)
Downloading sqltrie-0.11.2-py3-none-any.whl (17 kB)
Downloading appdirs-1.4.4-py2.py3-none-any.whl (9.6 kB)
Installing collected packages: pygtrie, funcy, appdirs, zc.lockfile, voluptuous, vine, sqltrie, shtab,
Successfully installed aiohttp-retry-2.9.1 amqp-5.3.1 appdirs-1.4.4 asyncssh-2.24.0 billiard-4.2.4 cel
```

Step 6: Initialize Git

`!git init`

Expected Output

Initialized empty Git repository



The screenshot shows a Jupyter Notebook cell with the command `!git init` and its output. The output includes several hints from Git and a confirmation message. The output is as follows:

```
[6]
✓ 0s !git init
...
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /content/drive/MyDrive/DVC_Project/.git/
```

Step 7: Initialize DVC

`!dvc init`

Expected Output

Initialized DVC repository

```
[7] !dvc init
✓ 1s
... Initialized DVC repository.

You can now commit the changes to git.

+-----+
| DVC has enabled anonymous aggregate usage analytics. |
| Read the analytics documentation (and how to opt-out) here: |
| <https://dvc.org/doc/user-guide/analytics> |
+-----+

What's next?
-----
- Check out the documentation: <https://dvc.org/doc>
- Get help and share ideas: <https://dvc.org/chat>
- Star us on GitHub: <https://github.com/treeverse/dvc>
```

Step 8: Create Dataset

import pandas as pd

```
data = {
    "Student":["Ravi","Priya","Anu","Rahul"],
    "Marks":[90,85,78,95]
}
```

```
df = pd.DataFrame(data)
```

```
df.to_csv("student.csv",index=False)
```

```
df.head()
```

```
import pandas as pd

data = {
    "Student": ["Ravi", "Priya", "Anu", "Rahul"],
    "Marks": [90, 85, 78, 95]
}

df = pd.DataFrame(data)

df.to_csv("student.csv", index=False)

df.head()
```

	Student	Marks
0	Ravi	90
1	Priya	85
2	Anu	78
3	Rahul	95

Next steps: [Generate code with df](#) [New interactive sheet](#)

Step 9: Add Dataset into DVC

`!dvc add student.csv`

Expected Output

student.csv.dvc created successfully

```
!dvc add student.csv

... Checking graph
Adding...: 0% 0/1 [00:00<?, ?file/s{'info': ''}]
!
    |0.00 [00:00, ?file/s]

!
    0% |      |0/? [00:00<?, ?files/s]

Adding student.csv to cache: 0% 0/1 [00:00<?, ?file/s]
Adding student.csv to cache: 0% 0/1 [00:00<?, ?file/s{'info': ''}]

    0% 0/1 [00:00<?, ?files/s]
    0% 0/1 [00:00<?, ?files/s{'info': ''}]
Adding...: 100% 1/1 [00:00<00:00, 3.45file/s{'info': ''}]

To track the changes with git, run:

    git add .gitignore student.csv.dvc

To enable auto staging, run:

    dvc config core.autostage true
```

Step 10: Check Files

`!ls`

Output

student.csv

student.csv.dvc

```
!ls
```

```
... student.csv student.csv.dvc
```

Step 11: Commit Version 1

!git config --global user.name "Your Name"

!git config --global user.email "your_email@example.com"

```
[1] ✓ 0s !git config --global user.name "Sirisha Potluri"
!git config --global user.email "sirisha.vegunta@gmail.com"
```

!git config --global --list

```
[2] ✓ 0s !git config --global --list
```

```
... user.name=Sirisha Potluri
user.email=sirisha.vegunta@gmail.com
```

from google.colab import drive

drive.mount('/content/drive')

```
from google.colab import drive
drive.mount('/content/drive')
```

```
... Mounted at /content/drive
```

%cd /content/drive/MyDrive

!ls

```

... /content/drive/MyDrive
'18-Payment receipt.pdf'
'20210722_151007.jpg'
'343-payment proof.jpeg'
'8863_DR. SIRISHA POTLURI_APR_2025 (1).pdf'
'Colab Notebooks'
'CV and Passport pics'
'Dr.Sirisha CV.docx'
'Dr.Sirisha Potluri_Short_CV_Latest.pdf'
DVC_Project
'Google Earth'
'ID cards'
'IEEE_Endorsement Letter.pdf'
'image (1).png'
'Land House Documents'
P-15
P-18
P-21
P-23
'Paper Id-4'
'Paper Id-4 (1)'
'Paper Id 97.pdf'
payment1.pdf
Payment-2.pdf
'payment proof_172.pdf'
'Payment receipt Id-19.pdf'
'Test file for chapter.docx'
'unamedmy pic.png'
'Vit FDP.pdf'
'WhatsApp Image 2024-07-04 at 9.26.52 PM.jpeg'

```

%cd /content/drive/MyDrive/DVC_Project

!ls -la

```

[8]
✓ 0s
%cd /content/drive/MyDrive/DVC_Project
!ls -la

... /content/drive/MyDrive/DVC_Project
total 10
drwx----- 2 root root 4096 Jul 22 10:26 .dvc
-rw----- 1 root root 139 Jul 22 10:26 .dvcignore
drwx----- 2 root root 4096 Jul 22 10:25 .git
-rw----- 1 root root 13 Jul 22 10:27 .gitignore
-rw----- 1 root root 47 Jul 22 10:28 student.csv
-rw----- 1 root root 89 Jul 22 10:28 student.csv.dvc

```

!git status

```

[9]
✓ 2s
!git status

... Refresh index: 100% (5/5), done.
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   .dvc/.gitignore
    new file:   .dvc/config
    new file:   .dvcignore
    new file:   .gitignore
    new file:   student.csv.dvc

```

!git add .

!git commit -m "Dataset Version 1"

```
[10]
✓ 1s !git add .
!git commit -m "Dataset Version 1"

... [master (root-commit) 141cdcf] Dataset Version 1
5 files changed, 12 insertions(+)
create mode 100644 .dvc/.gitignore
create mode 100644 .dvc/config
create mode 100644 .dvcignore
create mode 100644 .gitignore
create mode 100644 student.csv.dvc
```

Step 12: Modify Dataset

import pandas as pd

```
df = pd.read_csv("student.csv")
```

```
df.loc[len(df)] = ["Sita", 88]
```

```
df.to_csv("student.csv", index=False)
```

```
df
```

```
! import pandas as pd

df = pd.read_csv("student.csv")

df.loc[len(df)] = ["Sirisha", 88]

df.to_csv("student.csv", index=False)

df
```

...	Student	Marks	
0	Ravi	90	
1	Priya	85	
2	Anu	78	
3	Rahul	95	
4	Sirisha	88	

Step 13: Track New Version

```
!pip install dvc
```

```
!dvc version
```



!dvc version

```
... DVC version: 3.67.1 (pip)
-----
Platform: Python 3.12.13 on Linux-6.6.122+-x86_64-with-glibc2.35
Subprojects:
    dvc_data = 3.18.3
    dvc_objects = 5.2.0
    dvc_render = 1.0.2
    dvc_task = 0.40.2
    scmrepo = 3.6.2
Supports:
    http (aiohttp = 3.14.1, aiohttp-retry = 2.9.1),
    https (aiohttp = 3.14.1, aiohttp-retry = 2.9.1)
Config:
    Global: /root/.config/dvc
    System: /etc/xdg/dvc
Cache types:
Cache directory: fuse.drive on drive
Caches: local
Remotes: None
Workspace directory: fuse.drive on drive
Repo: dvc, git
Repo.site_cache_dir: /var/tmp/dvc/repo/103f29b302a97e755c0cfb63160e44ee
```

%cd /content/drive/MyDrive/DVC_Project



%cd /content/drive/MyDrive/DVC_Project

... /content/drive/MyDrive/DVC_Project

!dvc add student.csv



!dvc add student.csv

```
... * Checking graph
Adding...: 0% 0/1 [00:00<?, ?file/s{'info': ''}]
!
    |0.00 [00:00, ?file/s]

!
    0% |          |0/? [00:00<?, ?files/s]

Adding student.csv to cache: 0% 0/1 [00:00<?, ?file/s]
Adding student.csv to cache: 0% 0/1 [00:00<?, ?file/s{'info': ''}]

    0% 0/1 [00:00<?, ?files/s]
    0% 0/1 [00:00<?, ?files/s{'info': ''}]
Adding...: 100% 1/1 [00:00<00:00, 1.94file/s{'info': ''}]

To track the changes with git, run:

    git add student.csv.dvc

To enable auto staging, run:

    dvc config core.autostage true
```

Step 14: Commit Version 2

!git add .

!git commit -m "Dataset Version 2"

```
!git add .

!git commit -m "Dataset Version 2"

... [master 016d866] Dataset Version 2
1 file changed, 2 insertions(+), 2 deletions(-)
```

Step 15: View Version History

!git log --oneline

```
[18] !git log --oneline
✓ Os
... 016d866 (HEAD -> master) Dataset Version 2
141cdf Dataset Version 1
```

Step 16: Restore Previous Version

!git checkout HEAD~1

```
!git checkout HEAD~1

... Note: switching to 'HEAD~1'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

    git switch -c <new-branch-name>

Or undo this operation with:

    git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at 141cdf Dataset Version 1
```

!dvc checkout

```
!dvc checkout

... Building workspace index |1.00 [00:00, 20.9entry/s]
Comparing indexes |2.00 [00:00, 1.72kentry/s]
Applying changes |1.00 [00:00, 4.23file/s]
M student.csv
```

Step 17: Verify Dataset

import pandas as pd

pd.read_csv("student.csv")

Output

Version 1 Dataset Restored

↑

↓

:

▶

```
import pandas as pd
pd.read_csv("student.csv")
```

...

	Student	Marks
0	Ravi	90
1	Priya	85
2	Anu	78
3	Rahul	95